

Bull Spreads Using Call Options

AN INTRODUCTION TO CALL SPREAD STRATEGIES

A *spread* is a transaction in which one simultaneously buys one option and sells another option, with different terms, on the same underlying security. In a call spread, the options are all calls. The basic idea behind spreading is that the strategist is using the sale of one call to reduce the risk of buying another call. The short call in a spread is considered covered, for margin purposes, only if the long call has an expiration date equal to or longer than the short call. Before delving into the individual types of spreads, it may be beneficial to cover some general facts that pertain to most spread situations.

All spreads fall into three broad categories: *vertical*, *horizontal*, or *diagonal*. A *vertical spread* is one in which the calls involved have the same expiration date but different striking prices. An example might be to buy the XYZ October 30 and sell the October 35 simultaneously. A *horizontal spread* is one in which the calls have the same striking price but different expiration dates. This is a horizontal spread: Sell the XYZ January 35 and buy the XYZ April 35. A *diagonal spread* is any combination of vertical and horizontal and may involve calls that have different expiration dates as well as different striking prices. These three names that classify the spreads can be related to the way option prices are listed in any newspaper summary of closing option prices. A vertical spread involves two options from the same column in a newspaper listing. Newspaper columns run vertically. A horizontal spread involves two calls whose prices are listed in the same row in a newspaper listing; rows are horizontal. This relationship to the listing format in newspapers is not important, but it is an easy way to remember what vertical spreads and

horizontal spreads are. There are many types of vertical and horizontal spreads, and several of them are discussed in detail in later chapters.

SPREAD ORDER

The term “spread” designates not only a type of strategy, but a type of order as well. *All spread transactions in which both sides of the spread are opening (initial) transactions must be done in a margin account.* This means that the customer must generally maintain a minimum equity in the account, normally \$2,000. Some brokerage houses may also have a maintenance requirement, or “kicker.”

It is possible to transact a spread in a cash account, but one of the sides must be a closing transaction. In fact, many of the follow-up actions taken in the covered writing strategy are actually spread transactions. Suppose a covered writer is currently short one XYZ April call against 100 shares of the underlying stock. If he wants to roll forward to the July 35 call, he will be buying back the April 35 and selling the July 35 simultaneously. This is a spread transaction, technically, since one call is being bought and the other is being sold. However, in this transaction, the buy side is a closing transaction and the sell side is an opening transaction. This type of spread could be done in a cash account. Whenever a covered writer is rolling—up, down, or forward—he should place the order as a spread order to facilitate a better price execution.

The spreads discussed in the following chapters are predominantly spread strategies, ones in which both sides of the spread are opening transactions. These are designed to have their own profit and risk potentials, and are not merely follow-up actions to some previously discussed strategy.

When a spread order is entered, the options being bought and sold must be specified. Two other items must be specified as well: the price at which the spread is to be executed, and whether that price is a credit or a debit. If the total price of the spread results in a cash inflow to the spread strategist, the spread is a *credit spread*. This merely means that the sell side of the spread brings in a higher price than is paid for the buy side of the spread. If the reverse is true—that is, there is a cash outflow from the spread transaction—the spread is said to be a *debit spread*. This means that the buy side of the spread costs more than is received from the sell side. It is also common to refer to the purchased side of the spread as the *long side* and to refer to the written side of the spread as the *short side*.

The price at which a certain spread can be executed is generally *not* the difference between the last sale prices of the two options involved in the spread.

Example: An investor wants to buy an XYZ October 30 and simultaneously sell an XYZ October 35 call. If the last sale price of the October 30 was 4 points and the last sale price

of the October 35 was 2 points, it does not necessarily mean that the spread could be done for a 2-point debit (the difference in the last sale prices). In fact, *the only way to determine the market price for a spread transaction is to know what the bid and asked prices of the options involved are*. Suppose the following quotes are available on these two calls:

	Bid	Asked	Last Sale
October 30 call	3.90	4.10	4.00
October 35 call	1.95	2.00	2.00

Since the spread in question involves buying the October 30 call and selling the October 35, the spreader will, at market, have to pay 4.10 for the October 30 (the asked or offering quote) and will receive only 1.95 (the bid quote) for the October 35. This results in a debit of 2.15 points, significantly more than the 2-point difference in the last sale prices. Of course, one is free to specify any price he wants for any type of transaction. One might enter this spread order at a 2.10-point debit and could have a reasonable chance of having the order filled if the executing broker can do better than the bid side on the October 35 or better than the offering side on the October 30.

The point to be learned here is that *one cannot assume that last sale prices are indicative of the price at which a spread transaction can be executed*. This makes computer analysis of spread transactions via closing price data somewhat difficult. Some computer data services offer (generally at a higher cost) closing bid and asked prices as well as closing sale prices. If a strategist is forced to operate with closing prices only, however, he should attempt to build some screens into his output to allow for the fact that last sale prices might not be indicative of the price at which the spread can be executed. One simple method for screening is to look only at relatively liquid options—that is, those that have traded a substantial number of contracts during the previous trading day. If an option is experiencing a great deal of trading activity, there is a much better chance that the current quote is “tight,” meaning that the bid and offering prices are quite close to the last sale price.

In the early days of listed options, it was somewhat common practice to “leg” into a spread. That is, the strategist would place separate buy and sell orders for the two transactions comprising his spread. As the listed markets have developed, adding depth and liquidity, *it is generally a poor idea to leg into a spread*. If the floor broker handling the transaction knows the entire transaction, he has a much better chance of “splitting a quote,” buying on the bid, or selling on the offering. *Splitting a quote* merely means executing at a price that is between the current bid and asked prices. For example, if the bid is 3.90 and the offering is 4.10, a transaction at a price of 4 would be “splitting the quote.”

The public customer must be aware that spread transactions may involve substantially higher commission costs, because there are twice as many calls involved in any one transaction. Some brokers offer slightly lower rates for spread transactions, but these are not nearly as low as spreads in commodity trading, for example.

The *bull spread* is one of the most popular forms of spreading. In this type of spread, one buys a call at a certain striking price and sells a call at a higher striking price. Generally, both options have the same expiration date. This is a vertical spread. A *bull spread tends to be profitable if the underlying stock moves up in price; hence, it is a bullish position*. The spread has both limited profit potential and limited risk. Although both can be substantial percentagewise, the risk can never exceed the net investment. In fact, a bull spread requires a smaller dollar investment and therefore has a smaller maximum dollar loss potential than does an outright call purchase of a similar call.

Example: The following prices exist:

XYZ common, 32;

XYZ October 30 call, 3; and

XYZ October 35 call, 1.

A bull spread would be established by buying the October 30 call and simultaneously selling the October 35 call. Assume that this could be done at the indicated 2-point debit. A *call bull spread is always a debit transaction*, since the call with the lower striking price must always trade for more than a call with a higher price, if both have the same expiration date. Table 7-1 and Figure 7-1 depict the results of this transaction at expiration. The indicated call profits or losses would be realized if the calls were liquidated at parity at expiration. Note that *the spread has a maximum profit and this profit is realized if the stock is anywhere above the higher striking price at expiration*. The maximum loss is realized if the stock is anywhere below the lower strike at expiration, and is equal to the net investment, 2 points in this example.

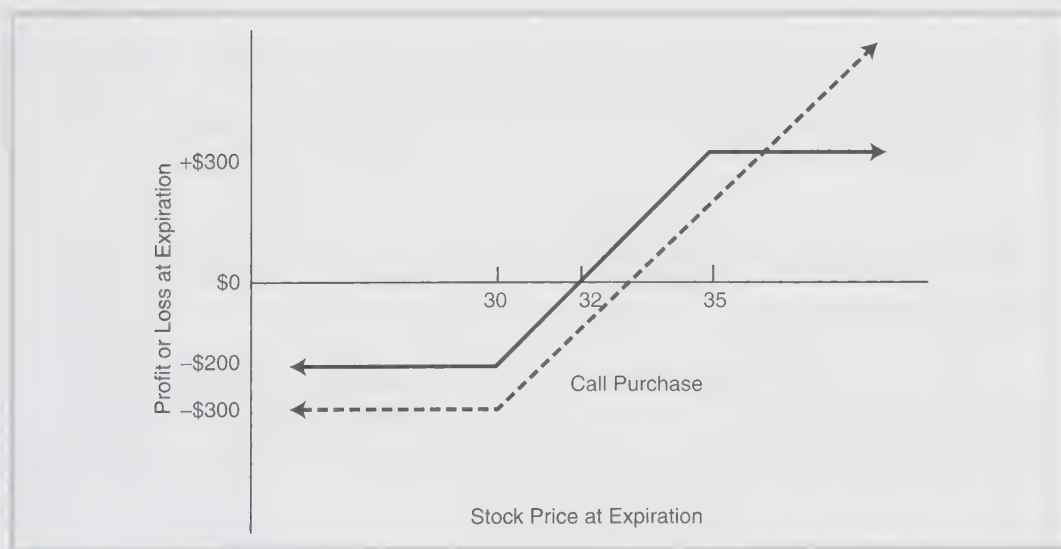
Moreover, there is a break-even point that always lies between the two striking prices at expiration. In this example, the break-even point is 32. All bull spreads have profit graphs with the same shape as the one shown in Figure 7-1 when the expiration dates are the same for both calls.

The investor who establishes this position is bullish on the underlying stock, but is generally looking for a way to hedge himself. If he were rampantly bullish, he would merely buy the October 30 call outright. However, the sale of the October 35 call against the purchase of the October 30 allows him to take a position that will outperform the outright purchase of the October 30, dollarwise, as long as the stock does not rise above 36 by expiration. This fact is demonstrated by the dashed line in Figure 7-1.

TABLE 7-1.
Results at expiration of bull spread.

XYZ Price at Expiration	October 30 Profit	October 35 Profit	Total Profit
25	− \$300	+\$100	−\$200
30	− 300	+ 100	− 200
32	− 100	+ 100	0
35	+ 200	+ 100	+ 300
40	+ 700	− 400	+ 300
45	+ 1,200	− 900	+ 300

FIGURE 7-1.
Bull spread.



Therefore, the strategist establishing the bull spread is bullish, but not overly so. To verify that this comparison is correct, note that if one bought the October 30 call outright for 3 points, he would have a 3-point profit at expiration if XYZ were at 36. Both strategies have a 3-point profit at 36 at expiration. Below 36, the bull spread does better because the sale of the October 35 call brings in the extra point of premium. Above 36 at expiration, the outright purchase outperforms the bull spread, because there is no limit on the profits that can occur in an outright purchase situation.

The net investment required for a bull spread is the net debit plus commissions. Since the spread must be transacted in a margin account, there will generally be a minimum

equity requirement imposed by the brokerage firm. In addition, there may be a maintenance requirement by some brokers. Suppose that one was establishing 10 spreads at the prices given in the example above. His investment, before commissions, would be \$2,000 (\$200 per spread), plus commissions. It is a simple matter to compute the break-even point and the maximum profit potential of a call bull spread:

$$\begin{array}{l} \text{Break-even point} = \text{Lower striking price} + \text{Net debit of spread} \\ \text{Maximum profit potential} = \text{Higher striking price} - \text{Lower striking price} - \text{Net debit of spread} \end{array}$$

In the example above, the net debit was 2 points. Therefore, the break-even point would be $30 + 2$, or 32. The maximum profit potential would be $35 - 30 - 2$, or 3 points. These figures agree with Table 7-1 and Figure 7-1. *Commissions may represent a significant percentage of the profit and net investment*, and should therefore be calculated before establishing the position. If these commissions are included in the net debit to establish the spread, they conveniently fit into the preceding formulae. Commission charges can be reduced percentagewise by spreading a large quantity of calls. For this reason, it is generally advisable to spread at least 5 options at a time.

DEGREES OF AGGRESSIVENESS

AGGRESSIVE BULL SPREAD

Depending on how the bull spread is constructed, it may be an extremely aggressive or more conservative position. The most commonly used bull spread is of the *aggressive* type; the stock is generally well below the higher striking price when the spread is established. This aggressive bull spread generally has the ability to generate substantial percentage returns if the underlying stock should rise in price far enough by expiration. *Aggressive bull spreads are most attractive when the underlying common stock is relatively close to the lower striking price at the time the spread is established.* A bull spread established under these conditions will generally be a low-cost spread with substantial profit potential, even after commissions are included.

EXTREMELY AGGRESSIVE BULL SPREAD

An extremely aggressive type of bull spread is the “out-of-the-money” spread. In such a spread, *both* calls are out-of-the-money when the spread is established. These spreads are extremely inexpensive to establish and have large potential profits if the stock should

climb to the higher striking price by expiration. However, they are usually quite deceptive in nature. The underlying stock has only a relatively remote chance of advancing such a great deal by expiration, and the spreader could realize a 100% loss of his investment even if the underlying stock advances moderately, since both calls are out-of-the-money. This spread is akin to buying a deeply out-of-the-money call as an outright speculation. It is not recommended that such a strategy be pursued with more than a very small percentage of one's speculative funds.

LEAST AGGRESSIVE BULL SPREAD

Another type of bull spread can be found occasionally—the “in-the-money” spread. In this situation, both calls are in-the-money. This is a much less aggressive position, since it offers a large probability of realizing the maximum profit potential, although that profit potential will be substantially smaller than the profit potentials offered by the more aggressive bull spreads.

Example: XYZ is at 37 some time before expiration, and the October 30 call is at 7 while the October 35 call is at 4. Both calls are in-the-money and the spread would cost 3 points (debit) to establish. The maximum profit potential is 2 points, but it would be realized as long as XYZ were above 35 at expiration. That is, XYZ could *fall* by 2 points and the spreader would still make his maximum profit. This is certainly a more conservative position than the aggressive spread described above. The commission costs in this spread would be substantially larger than those in the spreads above, which involve less expensive options initially, and they should therefore be figured into one's profit calculations before entering into the spread transaction. Since this stock would have to decline 7 points to fall below 30 and cause a loss of the entire investment, it would have to be considered a rather low-probability event. This fact adds to the less aggressive nature of this type of spread.

RANKING BULL SPREADS

To accurately compare the risk and reward potentials of the many bull spreads that are available in a given day, one has to use a computer to perform the mass calculations. It is possible to use a strictly arithmetic method of ranking bull spreads, but such a list will not be as accurate as the correct method of analysis. In reality, it is necessary to incorporate the volatility of the underlying stock, and possibly the expected return from the spread as well, into one's calculations. The concept of expected return is described in detail in Chapter 28, where a bull spread is used as an example.

The exact method for using volatility and predicting an option's price after an upward movement are presented later. Many data services offer such information. However, if the reader wants to attempt a simpler method of analysis, the following one may suffice. In any ranking of bull spreads, *it is important not to rank the spreads by their maximum potential profits at expiration*. Such a ranking will always give the most weight to deeply out-of-the-money spreads, which can rarely achieve their maximum profit potential. It would be better to screen out any spreads whose maximum profit prices are too far away from the current stock price. A simple method of allowing for a stock's movement might be to assume that the stock could, at expiration, advance by an amount equal to twice the time value premium in an at-the-money call. Since more volatile stocks have options with greater time value premium, this is a simple attempt to incorporate volatility into the analysis. Also, since longer-term options have more time value premium than do short-term options, this will allow for larger movements during a longer time period. Percentage returns should include commission costs. This simple analysis is not completely correct, but it may prove useful to those traders looking for a simple arithmetic method of analysis that can be computed quickly.

FURTHER CONSIDERATIONS

The bull spreads described in previous examples utilize the same expiration date for both the short call and the long call. It is sometimes useful to buy a call with a longer time to maturity than the short call has. Such a position is known as a diagonal bull spread and is discussed in a later chapter.

Experienced traders often turn to bull spreads when options are expensive. The sale of the option at the higher strike partially mitigates the cost of buying an expensive option at the lower strike. However, one should not always use the bull spread approach just because the options have a lot of time value premium, for he would be giving up a lot of upside profit potential in order to have a hedged position.

With most types of spreads, it is necessary for some time to pass for the spread to become significantly profitable, even if the underlying stock moves in favor of the spreader. For this reason, *bull spreads are not for traders* unless the options involved are very short-term in nature. If a speculator is bullishly oriented for a short-term upward move in an underlying stock, it is generally better for him to buy a call outright than to establish a bull spread. Since the spread differential changes mainly as a function of time, small movements in price by the underlying stock will not cause much of a short-term change in the price of the spread. However, the bull spread has a distinct advantage over the purchase of a call if the underlying stock advances moderately *by expiration*.

In the previous example, a bull spread was established by buying the XYZ October 30 call for 3 points and simultaneously selling the October 35 call for 1 point. This spread

can be compared to the outright purchase of the XYZ October 30 alone. There is a short-term advantage in using the outright purchase.

Example: The underlying stock jumps from 32 to 35 in one day's time. The October 30 would be selling for approximately 5.50 if that happened, and the outright purchaser would be ahead by 2.50 points, less one option commission. The long side of the bull spread would do as well, of course, since it utilizes the same option, but the short side, the October 35, would probably be selling for about 2.50. Thus, the bull spread would be worth 3 points in total (5.50 on the long side, less 2.50 loss on the short side). This represents a 1-point profit to the spreader, less two option commissions, since the spread was initially established at a debit of 2 points. Clearly, then, for the shortest time period—one day—the outright purchase outperforms the bull spread on a quick rise.

For a slightly longer time period, such as 30 days, the outright purchase still has the advantage if the underlying stock moves up quickly. Even if the stock should advance above 35 in 30 days, the bull spread will still have time premium in it and thus will not yet have reached its maximum spread potential of 5 points. Recall that the maximum potential of a bull spread is always equal to the difference between the striking prices. Clearly, the outright purchaser will do very well if the underlying stock should advance that far in 30 days' time. When risk is considered, however, it must be pointed out that the bull spread has fewer dollars at risk and, if the underlying stock should drop rather than rise, the bull spread will often have a smaller loss than the outright call purchase would.

The longer it takes for the underlying stock to advance, the more the advantage swings to the spread. Suppose XYZ does not get to 35 until expiration. In this case, the October 30 call would be worth 5 points and the October 35 call would be worthless. The outright purchase of the October 30 call would make a 2-point profit less one commission, but the spread would now have a 3-point profit, less two commissions. Even with the increased commissions, the spreader will make more of a profit, both dollarwise and percentagewise.

Many traders are disappointed with the low profits available from a bull spread when the stock rises almost immediately after the position is established. One way to partially offset the problem with the spread not widening out right away is to use a greater distance between the two strikes. When the distance is great, the spread has room to widen out, even though it won't reach its maximum profit potential right away. Still, since the strikes are "far apart," there is more room for the spread to widen even if the underlying stock rises immediately.

The conclusion that can be drawn from these examples is that, in general, the outright purchase is a better strategy if one is looking for a quick rise by the underlying stock. Overall, the bull spread is a less aggressive strategy than the outright purchase of a call.

The spread will not produce as much of a profit on a short-term move, or on a sustained, large upward move. It will, however, outperform the outright purchase of a call if the stock advances slowly and moderately by expiration. Also, the spread always involves fewer actual dollars of risk, because it requires a smaller debit to establish initially. Table 7-2 summarizes which strategy has the upper hand for various stock movements over differing time periods.

TABLE 7-2.**Bull spread and outright purchase compared.**

		If the underlying stock . . .		
	Declines	Remains Relatively Unchanged	Advances Moderately	Advances Substantially
in . . .				
1 week	Bull spread	Bull spread	Outright purchase	Outright purchase
1 month	Bull spread	Bull spread	Outright purchase	Outright purchase
At expiration	Bull spread	Bull spread	Bull spread	Outright purchase

FOLLOW-UP ACTION

Since the strategy has both limited profit and limited risk, it is not mandatory for the spreader to take any follow-up action prior to expiration. If the underlying stock advances substantially, the spreader should watch the time value premium in the short call closely in order to close the spread if it appears that there is a possibility of assignment. This possibility would increase substantially if the time value premium disappeared from the short call. If the stock falls, the trader may want to close the spread in order to limit his losses even further.

When the spread is closed, the order should also be entered as a spread transaction. If the underlying stock has moved up in price, the order to liquidate would be a *credit* spread involving two closing transaction. *The maximum credit that can be recovered from a bull spread is an amount equal to the difference between the striking prices.* In the previous example, if XYZ were above 35 at expiration, one might enter an order to liquidate the spread as follows: Buy the October 35 (it is common practice to specify the buy side of a spread first when placing an order); sell the October 30 at a 5-point credit. In reality, because of the difference between bids and offers, it is quite difficult to obtain the entire 5-point credit even if expiration is quite near. Generally, one might ask for a 4.80 or 4.90 credit. It is possible to close the spread via exercise, although this method is normally advisable only for traders who pay little or no commissions. If the short side of a

spread is assigned, the spreader may satisfy the assignment notice by exercising the long side of his spread. There is no margin required to do so, but there are stock commissions involved. Since these stock commissions to a public customer would be substantially larger than the option commissions involved in closing the spread by liquidating the options, *it is recommended that the public customer attempt to liquidate rather than exercise.*

A minor point should be made here. Since the amount of commissions paid to liquidate the spread would be larger if higher call prices are involved, the actual net maximum profit point for a bull spread is for the stock to be exactly at the higher striking price at expiration. If the stock exceeds the higher striking price by a great deal, the gross profit will be the same (it was demonstrated earlier that this gross profit is the same anywhere above the higher strike at expiration), but the net profit will be slightly smaller, since the investor will pay more in commissions to liquidate the spread.

Some spreaders prefer to buy back the short call if the underlying stock drops in price, in order to lock in the profit on the short side. They will then hold the long call in hopes of a rise in price by the underlying stock, in order to make the long side of the spread profitable as well. This amounts to “legging” out of the spread, although the overall increase in risk is small—the amount paid to repurchase the short call. If he attempts to “leg” out of the spread in such a manner, the spreader should not attempt to buy back the short call at too high a price. If it can be repurchased at $\frac{1}{8}$; or $\frac{1}{16}$, the spreader will be giving away virtually nothing by buying back the short call. However, he should not be quick to repurchase it if it still has much more value than that, unless he is closing out the entire spread. At no time should one attempt to “leg” out after a stock price increase, taking the profit on the long side and hoping for a stock price decline to make the short side profitable as well. The risk is too great.

Many traders find themselves in the somewhat perplexing situation of having seen the underlying make a large, quick move, only to find that their spread has not widened out much. They often try to figure out a way to perhaps lock in some gains in case the underlying subsequently drops in price, but they want to be able to wait around for the spread to widen out more toward its maximum profit potential. There really isn't any hedge that can accomplish all of these things. The only position that can lock in the profits in a call bull spread is to purchase the accompanying *put* bear spread. This strategy is discussed in Chapter 23, Spreads Combining Calls and Puts.

OTHER USES OF BULL SPREADS

Superficially, the bull spread is one of the simplest forms of spreading. However, it can be an extremely useful tool in a wide variety of situations. Two such situations were described in Chapter 3. If the outright purchaser of a call finds himself with an unrealized loss, he

may be able to substantially improve his chances of getting out even by “rolling down” into a bull spread. If, however, he has an unrealized profit, he may be able to sell a call at the next higher strike, creating a bull spread, in an attempt to lock in some of his profit.

In a somewhat similar manner, a common stockholder who is faced with an unrealized loss may be able to utilize a bull spread to lower the price at which he can break even. He may often have a significantly better chance of breaking even or making a profit by using options. The following example illustrates the stockholder’s strategy.

Example: An investor buys 100 shares of XYZ at 48, and later finds himself with an unrealized loss with the stock at 42. A 6-point rally in the stock would be necessary in order to break even. However, if XYZ has listed options trading, he may be able to significantly reduce his break-even price. The prices are:

XYZ common, 42;

XYZ October 40, 4; and

XYZ October 45, 2.

The stock owner could enhance his overall position by buying one October 40 call and selling *two* October 45 calls. Note that no extra money, except commissions, is required for this transaction, because the credit received from selling two October 45’s is \$400 and is equal to the cost of buying the October 40 call. However, maintenance and equity requirements still apply, because a spread has been established.

The resulting position does not have an uncovered, or naked, option in it. One of the October 45 calls that was sold is covered by the underlying stock itself. The other is part of a bull spread with the October 40 call. It is not particularly important that the resulting position is a combination of both a bull spread and a covered write. What is important is the profit characteristic of this new total position.

If XYZ should continue to decline in price and be below 40 at October expiration, all the calls will expire worthless, and the resulting loss to the stock owner will be the same (except for the option commissions spent) as if he had merely held onto his stock without having done any option trading.

Since both a covered write and a bull spread are strategies with limited profit potential, *this new position obviously must have a limited profit*. If XYZ is anywhere above 45 at October expiration, the maximum profit will be realized. To determine the size of the maximum profit, assume that XYZ is at exactly 45 at expiration. In that case, the two short October 45’s would expire worthless and the long October 40 call would be worth 5 points. The option trades would have resulted in a \$400 profit on the short side (\$200 from each October 45 call) plus a \$100 profit on the long side, for a total profit of \$500 from the

option trades. Since the stock was originally bought at 48 in this example, the stock portion of the position is a \$300 loss with XYZ at 45 at expiration. The overall profit of the position is thus \$500 less \$300, or \$200.

For stock prices between 40 and 45 at expiration, the results are shown in Table 7-3 and Figure 7-2. Figure 7-2 depicts the two columns from the table labeled “Profit on Stock” and “Total Profit,” so that one can visualize how the new total position compares with the original stockholder’s profit. Several points should be noted from either the graph or the table. First, the break-even point is lowered from 48 to 44. The new total position breaks even at 44, so that only a 2-point rally by the stock by expiration is necessary in order to break even. The two strategies are equal at 50 at expiration. That is, the stock would have to rally more than 8 points, from 42 to 50, by expiration for the original stockholder’s position to outperform the new position. Below 40, the two strategies produce the same result. Finally, between 40 and 50, the new position outperforms the original stockholder’s position.

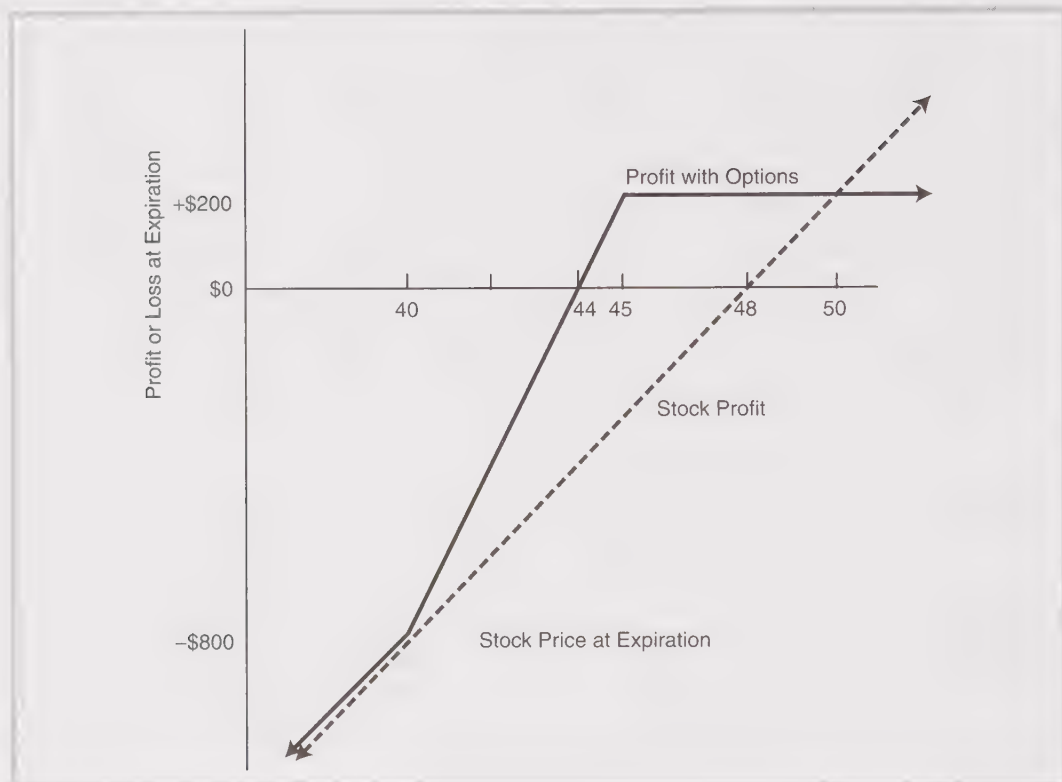
In summary, then, the stockholder stands to gain much and gives away very little by adding the indicated options to his stock position. If the stock stabilizes at all—anywhere between 40 and 50 in the example above—the new position would be an improvement. Moreover, the investor can break even or make profits on a small rally. If the stock continues to drop heavily, nothing additional will be lost except for option commissions. Only if the stock rallies very sharply will the stock position outperform the total position.

This strategy—combining a covered write and a bull spread—is sometimes used as an initial (opening) trade as well. That is, an investor who is considering buying XYZ at 42

TABLE 7-3.
Lowering the break-even price on common stock.

XYZ Price at Expiration	Profit on Stock	Profit on Short October 45’s	Profit on Long October 40	Total Profit
35	–\$1,300	+\$400	–\$400	–\$1,300
38	– 1,000	+ 400	– 400	– 1,000
40	– 800	+ 400	– 400	– 800
42	– 600	+ 400	– 200	– 400
43	– 500	+ 400	– 100	– 200
44	– 400	+ 400	0	0
45	– 300	+ 400	+ 100	+ 200
48	0	– 200	+ 400	+ 200
50	+ 200	– 600	+ 600	+ 200

FIGURE 7-2.
Lowering the break-even price on common stock.



might decide to buy the October 40 and sell two October 45's (for even money) at the outset. The resulting position would not be inferior to the outright purchase of XYZ stock, in terms of profit potential, unless XYZ rose above 46 by October expiration.

Bull spreads may also be used as a "substitute" for covered writing. Recall from Chapter 2 that writing against warrants can be useful because of the smaller investment required, especially if the warrant was in-the-money and was not selling at much of a premium. The same thinking applies to call options. *If there is an in-the-money call with little or no time premium remaining in it, its purchase may be used as a substitute for buying the stock itself.* Of course, the call will expire, whereas the stock will not; but the profit potential of owning a deeply in-the-money call can be very similar to owning the stock. Since such a call costs less to purchase than the stock itself would, the buyer is getting essentially the same profit or loss potential with a smaller investment. It is natural, then, to think that one might write another call—one closer to the money—against the

deeply in-the-money purchased call. This position would have profit characteristics much like a covered write, since the long call “simulates” the purchase of stock. This position really is, of course, a bull spread, in which the purchased call is well in-the-money and the written call is closer to the money. Clearly, one would not want to put all of his money into such a strategy and forsake covered writing, since, with bull spreads, he could be entirely wiped out in a moderate market decline. In a covered writing strategy, one still owns the stocks even after a severe market decline. However, one may achieve something of a compromise by investing a much smaller amount of money in bull spreads than he might have invested in covered writes. He can still retain the same profit potential. The balance of the investor’s funds could then be placed in interest-bearing securities.

Example: The following prices exist:

XYZ common, 49;
 XYZ April 50 call, 3; and
 XYZ April 35 call, 14.

Since the deeply in-the-money call has no time premium, its purchase will perform much like the purchase of the stock until April expiration. Table 7-4 summarizes the profit potential from the covered write or the bull spread. The profit potentials are the same from a cash covered write or the bull spread. Both would yield a \$400 profit before commissions if XYZ were above 50 at April expiration. However, since the bull spread requires a much smaller investment, the spreader could put \$3,500 into interest-bearing securities. This interest could be considered the equivalent of receiving the dividends on the stock. In any case, the spreader can lose only \$1,100, even if the stock declines substantially. The covered writer could have a larger unrealized loss than that if XYZ were below 35 at expiration. Also, in the bull spread situation, the writer can “roll down” the April 50 call if the stock declines in price, just as he might do in a covered writing situation.

TABLE 7-4.
Results for covered write and bull spread compared.

	Covered Write: Buy XYZ and Sell April 50 Call	Bull Spread: Buy XYZ April 35 Call and Sell April 50 Call
Maximum profit potential (stock over 50 in April)	\$ 400	\$ 400
break-even point	46	46
Investment	\$4,600	\$1,100

Thus, the bull spread offers the same dollar rewards, the same break-even point, smaller commission costs, less potential risk, and interest income from the fixed-income portion of the investment. While it is not always possible to find a deeply in-the-money call to use as a “substitute” for buying the stock, when one does exist, the strategist should consider using the bull spread instead of the covered write.

SUMMARY

The bull spread is one of the simplest and most popular forms of spreading. It will generally perform best in a moderately bullish environment. A bull spread will not widen out to its maximum profit potential right away, though; so for short-term trades, the outright purchase of a call is a better choice. The bull spread can also be applied for more sophisticated purposes in a far wider range of situations than merely wanting to attempt to capitalize on a moderate advance by the underlying stock. Both call buyers and stock buyers may be able to use bull spreads to “roll down” and produce lower break-even points for their positions. The covered writer may also be able to use bull spreads as a substitute for covered writes in certain situations in which a deeply in-the-money call exists.